Georgy Rached

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EDUCATION

Northeastern University | Boston, MA May 2029
Bachelor of Science in Electrical Engineering GPA: 3.96
Relevant Coursework: Circuits and Signals, Embedded Design, and Computing Fundamentals Dean's List

TECHNICAL SKILLS

CAD & Software: KiCad, Altium, LTspice, AutoCAD, FPGA/Quartus, SolidWorks, Git/GitHub
Programming & Microcontrollers: Python, C/C++, MATLAB, Arduino, Raspberry Pi Pico, ESP32
Tools & Manufacturing: Soldering, Multimeter, Oscilloscope, Waveform Generator, Circuit Analysis, PCB Design
Training & Certifications: Altium Education Course, Core MATLAB Skills

ENGINEERING PROJECTS

RP2350 Development Board (Personal Project)

KiCad, DFM/DFA, RP2350 July - August 2026

- Designed and brought up a 4-layer RP2350 development board per Raspberry Pi's design guide: 12 MHz crystal, QSPI boot flash, on-chip switching regulator, 3.3 V LDO, USB-C port with ESD protection, and additional components.
- Moved from 2 to 4 layers for a clean MCU via escape and an uninterrupted ground plane under the USB pair and crystal, trading fab cost for signal integrity.
- Re-sourced the bill of materials to in-stock basic passives after an initial quote came back inflated with minimum-order reels and special part fees, cutting assembly cost by 50%.

Room Temperature Monitor PCB (Personal Project)

Altium, Arduino Nano September 2026

- Designed and laid out a custom thermometer-shaped Altium board for an Arduino Nano room temperature monitor, driving three threshold LEDs off an LM34AH sensor.
- Thermally isolated the LM34 with separate ground pours under the Nano and the sensor, bridged by a copper-free neck, so the board's self-heating couldn't skew readings.

Blinking LED Heart PCB (Personal Project)

KiCad, 555 timer August 2025 - June 2026

- Designed a heart-shaped two-layer PCB driving 14 LEDs from a 555 timer with a potentiometer for variable blink speed; validated in TinkerCAD, then schematic capture, part selection, and routing in KiCad.
- Pivoted from a 9 V battery to a single 3 V coin cell using a low-voltage CMOS 555 and 2 V LEDs, balancing coin cell life and LED brightness.

Frog Launcher (Course Project)

Raspberry Pi Pico, MicroPython, LCD October - December 2025

- Led software for a four-person embedded arcade game on a Raspberry Pi Pico, extending into CAD correction and laser fabrication of all structural parts.
- Rewired and debugged the entire button/LCD circuit after half the inputs failed at prototype review, tracing faults through undocumented wiring and isolating an apparent Pico failure to excess LED current draw.

WORK EXPERIENCE

Crew Member

Dunkin' Donuts | Lawrence, MA January 2025 - July 2025

- Served 120+ customers daily through mobile orders, drive-thru, and counter orders while preparing food/drinks and maintaining store cleanliness.
- Trained 3 new hires on POS, food safety, and general procedures.